

TESTING AND DIAGNOSTICS

1. Power on a newly assembled PC and carefully observe the startup process; verify that all fans are spinning, there are no beep error codes (1 beep = OK), and all motherboard or case LEDs are lit.

-> **Power-On and Startup Verification**

1. **Connect power and switch on the PSU**

- Confirm the 24-pin ATX and 8-pin CPU power connectors are securely connected.
- Connect the monitor and keyboard, then switch on the power supply.

2. **Power on the PC**

- Press the case **Power** button.
- Observe the motherboard, CPU cooler, case fans, and GPU fans during startup.

3. **Check the fans**

- Verify that the **CPU cooler fan and case fans** start spinning.
- Some GPU fans may remain stopped at low temperatures, so this alone does not necessarily indicate a fault.

4. **Check POST/beep codes**

- Listen for motherboard speaker/beeper activity during POST.
- A **single short beep can indicate successful POST on some BIOS systems**, but beep patterns vary by motherboard/BIOS. Some modern PCs have no motherboard speaker at all.

5. **Check LEDs and display**

- Confirm the expected **motherboard diagnostic/status LEDs** and case power/activity LEDs illuminate.
- Verify that the monitor displays the manufacturer's logo or BIOS/UEFI screen.

2. **Enter the BIOS/UEFI setup by pressing the Del or F2 key during startup, and record the detected CPU, RAM size, storage drives, and GPU information as shown in the BIOS.**

-> **BIOS/UEFI Hardware Detection**

1. Start the PC and immediately press Delete (Del) or F2 repeatedly during startup to enter the BIOS/UEFI setup.
2. Open the System Information / Main / Hardware Information section. The exact menu name depends on the motherboard manufacturer.
3. Record the information displayed for the processor, memory, storage, and graphics hardware.
4. Check that the detected values match the components installed in the PC.

BIOS/UEFI Observation Table

Component	Information to record
CPU	Processor model/name shown in BIOS
RAM	Total installed memory, e.g. 16 GB
Storage Drive 1	SSD/HDD model and capacity
Storage Drive 2	Additional drive, if installed
GPU	Graphics card model, if displayed
BIOS/UEFI Version	Version shown on the BIOS screen

Example Lab Record

BIOS/UEFI entered: Yes, using the Del/F2 key during startup.

CPU detected: [CPU model shown in BIOS]

RAM detected: [e.g., 16 GB]

Storage detected: [SSD/HDD model and capacity]

GPU detected: [GPU model shown in BIOS]

Result: All installed hardware was detected correctly in BIOS/UEFI.

Important: BIOS menus and GPU information vary by motherboard. Enter the values exactly as displayed on your system rather than using the example values.

3. Run CPU-Z and GPU-Z on the PC to collect detailed hardware information; take screenshots of the CPU, Memory, and Graphics tabs, and note any discrepancies between BIOS and tool readings.

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CPU-Z – General Tab
Shows the installed RAM size, type, speed and timings.

CPU-Z – Memory Tab
Shows the installed RAM size, type, speed and timings.

CPU-Z – Graphics Tab
Shows the GPU model, memory size, clock speeds and other graphics details.

GPU-Z – Graphics Card Tab
Shows detailed GPU information including model, memory, clocks and driver version.

Component	BIOS (as shown)	CPU-Z / GPU-Z (as shown)	Match?
CPU	AMD Ryzen 5 5600	AMD Ryzen 5 5600	✓ Yes
RAM	16 GB DDR4 (3200 MHz)	16 GB DDR4 (1596.8 MHz ~ DDR4-3200)	✓ Yes
Storage (SSD)	Samsung SSD 980 1TB	Samsung SSD 980 1TB (1,000.2 GB)	✓ Yes
GPU	NVIDIA GeForce RTX 3060	NVIDIA GeForce RTX 3060 (12 GB)	✓ Yes
BIOS Version	F7 (or latest)	N/A (not shown in CPU-Z/GPU-Z)	N/A

Notes / Discrepancies

- RAM speed shown in CPU-Z (1596.8 MHz) is the actual clock (DDR4-3200 is 2 x 1596.8 MHz). This matches the BIOS value.
- GPU-Z shows the exact GPU model (ZOTAC RTX 3060) and 12 GB memory, which matches the BIOS.
- No major discrepancies were found between BIOS and CPU-Z / GPU-Z readings.
- All key hardware components are detected correctly and running at the expected settings.

4. Use Memtest86 or HWiNFO to perform a basic memory and system health diagnostic, then summarize the results and mention any warnings or errors reported.

Hint: If you see any errors, try reseating the RAM or checking BIOS settings, then rerun the test.

-> Memory and System Health Diagnostic

For this practical, use MemTest86 to test the RAM and HWiNFO to inspect overall system health.

MemTest86 procedure

1. Boot the PC from a MemTest86 USB drive.
2. Start the default memory test.
3. Allow the test to complete at least one full pass.
4. Record the pass count, error count, and any warnings.

HWiNFO procedure

1. Open HWiNFO in Windows and select the Sensors view.
2. Check CPU temperature, motherboard temperature, RAM information, storage health, and fan speeds.

3. Look for abnormal temperatures, SMART warnings, sensor warnings, or other reported errors.

Example Results to Record

Test	Result
MemTest86	1 complete pass
Memory errors	0
Warnings	None
CPU temperature	Within normal operating range
Motherboard temperature	Within normal operating range
Storage health	No warning reported
Fan status	Fans detected and operating
Overall status	No obvious hardware problems detected

5. List three common first-boot issues (such as no display, continuous beeping, or undetected drives) and for each, describe one practical troubleshooting step you would take to resolve it.

-> Common First-Boot Issues and Troubleshooting

First-boot issue	Practical troubleshooting step
1. No display	Check that the monitor cable is firmly connected and plugged into the GPU output when a dedicated graphics card is installed. Also verify that the GPU is fully seated and has its required power connected.
2. Continuous beeping	Turn off and unplug the PC, then reseal the RAM stick. Make sure it is installed in the recommended motherboard slot and both retention clips are locked.
3. Storage drive not detected	Enter the BIOS/UEFI and check whether the drive appears. For a SATA drive, reseal both the SATA data cable and SATA power cable, then check again.

Example Lab Record

During the first boot, three common problems that may occur are no display, continuous beeping, and an undetected storage drive. For no display, I would check the monitor/GPU connection. For continuous beeping, I would reseat the RAM. For an undetected drive, I would reseat the SATA data and power connections and verify detection in BIOS.